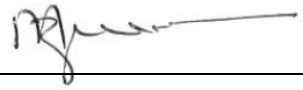


TRANSNET FREIGHT RAIL – TECHNOLOGY MANAGEMENT - TRAIN DESIGN TECHNOLOGIES

INSTRUCTION FOR THE RUNNING OF AIRBRAKE CONTAINER TRAINS NOT EXCEEDING 75 WAGONS AND/OR 3600 TONS BETWEEN CAPITAL PARK/KASERNE/CITY DEEP AND BAYHEAD/KINGSREST

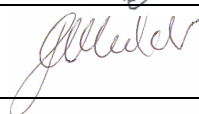
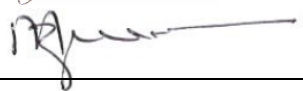
**Document Number :
S.GFB/TES/TE/OI 257**

Compiling officer:	A.R. Boy : 	Revision Number : 12
1st Technical Approving Officer	A.R. Boy : 	Issue date: 2016-08-15
2nd Technical Approving Officer	J.M. Mulder : 	Page Number : 1 of 26
Approving Safety Officer:	T. Zwane : 	



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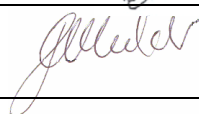
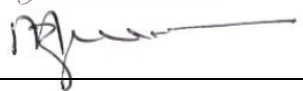
1 SCOPE

- 1.1** This document authorizes the running of Airbrake Container trains of up to 75 wagons or 3600 tons between Capital Park/Kaserne/City Deep and Bayhead/Kingsrest in both directions.
- 1.2** This document details the Technical Train Operating Instructions for the running of Airbrake Container Trains (As defined in Clause 3 below).
- 1.3** This revision is issued whereby the wagon compilation as specified in clause 9 as contained on page 17 of the instruction has been amended according to document with reference RSE/TE/LPR/0040 (Revision 2) dated 26 July 2016 and titled '*Loading Specification for Securing and Transporting of Containers*'.

2 ADDITIONAL INSTRUCTIONS

- 2.1** This document cancels and replaces any previous versions of this document as well as Z22/3 dated 15 November 1995 and Z103/7 dated 15 November 1995. In addition instruction with reference S.GFB/TES/TE/OI 0395 is superseded insofar as the specific route is concerned. Any previously issued Technical Train Operating Instructions pertinent to the specific train types as authorized and provided for within this authority that might exist are also hence rendered null and void.
- 2.2** Conditions contained in Circular X54 and General Appendix (Part I), not overruled by this instruction, must be adhered to.
- 2.3** Instruction S.GFB/TES/TE/OI0268 – General authorization for the haulage of dead locomotives on airbrake trains.



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3 GENERAL DEFINITIONS

3.1 Airbrake Container Train

An airbrake container train is a train where each individual wagon has an axle loading not exceeding twenty (20) tons per axle (t/a).

3.2 Empty Airbrake Container Train

An Empty Airbrake container train is a train consisting entirely of empty wagons.

4 TRAIN SPEEDS

4.1 Loaded and Empty Trains

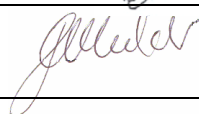
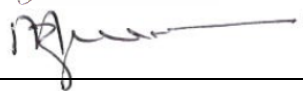
4.1.1 The maximum authorised operating train speed for all container train types is 60km/h subject to line speed, or unless a lower speed restriction has been enforced for any other reason whatsoever.

4.1.2 The speed of 60 km/h (as indicated in clause 4.1.1 above) up to an absolute maximum of 80 km/h may however only be exceeded for 'swing' purposes where the train handling method requires such speed for energy saving purposes.

Typically what is implied above is what is commonly referred to as 'Swing'. This may be described or defined as a situation whereby the authorized train speed of 60 km/h may be exceeded to an absolute maximum speed of 80 km/h on a down gradient in order to convert as much potential energy as possible into kinetic energy to facilitate energy saving where the train will shortly need to negotiate an ascending gradient. Under no circumstances however must the speed limit of the line be exceeded though. Energy wastage can sometimes not be avoided and only energy that can be kept within the speed limits can be retained.

The following may be used as a basic guideline relevant to train handling techniques regarding 'swing'.

From an energy conservation perspective when **moving from an ascending to a down gradient** it is desirable to move slower over the crest than the maximum line speed permits in order to use the downhill gradient forces to accelerate the train rather than the use of tractive power.

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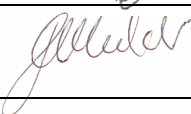
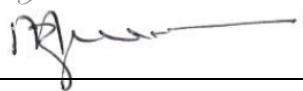
Every vehicle which moves over the crest will increase the tractive force on the vehicles still on the uphill. It is thus very important to reduce tractive power as soon as the first locomotive moves over the crest. Decide on an appropriate speed when negotiating the crest that will waste the least amount of energy on the downhill. Make use of the gradient forces to slow down the train by starting to reduce the tractive power at a point before the crest is reached. Allow the train to absorb the reduction in tractive power before reducing the power further. Monitor the changes continuously.

When the train has reached the planned speed, the locomotives must be on the crest and moving to the downhill. The tractive power can now be slowly reduced with care being taken not to let the train accelerate. A deceleration would be more acceptable than any acceleration. If the train would accelerate, then the front of the train is moving away from the rear and this can result in forces in the train that may damage the draw gears of the wagons on the crest.

While the train moves over the crest, the tractive power must slowly be reduced until the accelerator is totally closed. The ideal speed to go over the crest is between 25 and 28 km/h, depending on the line after the crest and the length of the downhill. The train driver may then allow the gradient force to accelerate the train. The speed chosen when cresting should be such that it would preferably not be necessary to apply brakes during this free run accelerating. Good knowledge of the line will aid in deciding on the speed, closing of the accelerator and the use of brakes. The steepness of the uphill and downhill together with the length and mass of the train must be kept in mind. It should be noted that if the down gradient is longer than 2 km and has a speed restriction applicable to it no 'swing' is permitted. In addition on any down gradient where train brakes need to be used to control the speed of the train then no 'swing' is permitted.

When **moving from a level to a down gradient** the train driver must plan carefully so as to maximise the use of the train's rolling resistance to reduce speed and allow the train to run at a realistic speed over the level and onto the downhill so that the maximum of potential energy may be converted to kinetic energy. The planning must be such that it is not necessary for the train driver to interfere with the natural movement of the train. Under no circumstances must the speed limit of the line be exceeded though.

When the train **moves from a downhill to an uphill** the train driver must plan ahead to carry over as much of the potential energy as possible by using the information on the line, signal positions and speed limits. Knowledge of the line is very important because the length of the downhill and uphill directly influences the train handling techniques to be deployed. According to knowledge of the line, a speed must be chosen whereby the maximum speed is not exceeded in the bottom of the dip. Allow the train to move up the uphill as far as possible by using only momentum. Closely observe the train speed and when the speed starts to fall below 60 km/h then

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gradually begin applying tractive power slowly so that a constant speed can be maintained. Allow enough time for the train to absorb the increase in tractive power before increasing the power further.

4.1.3 If the train length is less than 10 vehicles the absolute maximum train speed is restricted to 60 km/h.

4.1.4 Speed of trains on down gradients:

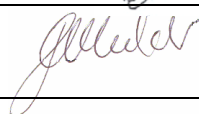
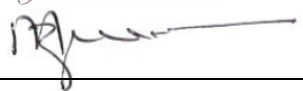
Taking clause 5 as contained below into cognisance and provided that track conditions and train composition permits and no other instructions or circumstances exist that authorise, make desirable or demand lower speeds, trains must observe the following speeds on down grades of 2 km and longer:

AVERAGE GRADIENT TRAIN SPEED

Gradient	Electric / Dynamic Brake	
	Sufficient	Insufficient
1 in 71 to 1 in 90	50km/h	35km/h
1 in 56 to 1 in 70	40km/h	25 km/h
1 in 55 and steeper	30km/h	20 km/h

4.1.5 The speed of 50 km/h may only be exceeded when it is within the capability of the locomotive/s to control the speed safely with electric brake only.



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5 DOWNGRADE SPEED RESTRICTIONS

- 5.1 Taking cognisance of paragraph 4 (above), trains should be operated utilising speeds as depicted hereunder as a guideline.

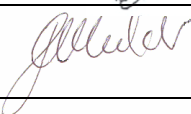
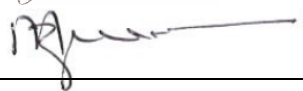
Kaserne/City Deep - Kaydale (both directions)

Note: * - Refers to Clause 4 above when sufficient Electric brake is not available

Downgrade speed restrictions		Station	Downgrade speed restrictions	
Speed km/h	km Point		km Point	Speed km/h
		Kaserne / City Deep	None	None
		Jupiter		
		India		
		Park Hill		
		Elsburg		
		Dallas		
		Wattles		
		Union		
50 or *35	0.5 to 3.0	Rooikop		
		Mapleton		
		Riet Vallei		
		Glenroy		
		Driemanskap		
		Kaydale		

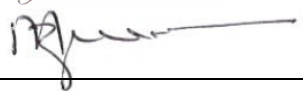
**INSTRUCTION FOR THE RUNNING OF AIRBRAKE CONTAINER
TRAINS NOT EXCEEDING 75 WAGONS AND/OR 3600 TONS
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Kaydale - Vooruitsig (both directions)

Downgrade speed restrictions		Station	Downgrade speed restrictions	
Speed km/h	km Point		km Point	Speed km/h
		Kaydale		
		Heidelberg		
		Kraal	43 - 52	50 / *35
50 or *35	58.0 to 62	Spruysrus		
		Fortuna		
		Balfour Noth	73 - 78	50 / *35
		Sprucewell		
50 or *35	97.5 to 99.5	Greylingstad		
		Teakworth		
		Beytelsplaat		
		Val		
		Cedarmont		
		Holmdene		
		Elmtree		
		Standerton	150 to 153	50 or *35
		Firham		
		Kromdraai		
		Kanonheuwel		
		Platrand	181 - 183	50 / *35
		Rusthof		
		Perdekop		
50 or *35	204.5 to 208.5	Palmford		
		Beechwick		
		Sandspruit		
		Vooruitsig		

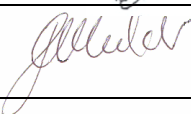
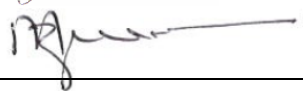
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Vooruitsig - Ladysmith (both directions)

Downgrade speed restrictions		Station	Downgrade speed restrictions	
Speed km/h	km Point		km Point	Speed km/h
40 or *25	0 to 4.5	Vooruitsig		
40 or *25	9.0 to 34	Clavis Laingsnek Mount Propect Glenharte Ingogo		
40 or *25	35.5 to 40			
40 or *25	54 to 58 68 to 70.5	Contarf Wykom Signal Hill Newcastle Ngagane Ballengeich	51.5 - 54 60 - 62.5	50 / *35 50 / *35
			85.5 to 90.0	50 or *35
			91.0 to 95.5	40 or *25
50 or *35	102.5 to 107.0	Danhauser		
30 or *20	121.5 to 130.5	Hattingspruit Glencoe		
40 or *25	130.5 to 136.0	Uithoek		
		Wasbank	141.5 to 147.5	50 or *35
40 / *25	148-150.5	Wesselsnek Elandslaagte Pepworth Danskraal North Danskraal South Ladysmith		
40 or *25	178 to 182.0			

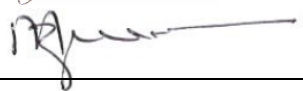
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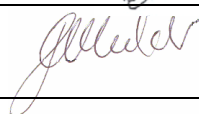
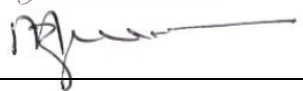
Ladysmith – Lions River (both directions)

Downgrade speed restrictions		Station	Downgrade speed restrictions	
Speed km/h	km Point		km Point	Speed km/h
		Ladysmith		
30 or *20	15.0 to 19.0	Umbulwana		
		Pieters	26 to 31	40 or *25
		Colenso	44 to 51	40 or *25
		Chieveley	68 to 80	40 or *25
		Frere	81 to 88.0	No. 2 Line 30 / *20 No. 1 Line 40 / *25
50 or *35	88.5 to 93	Ennersdale	99.0 to 103.0	40 or *25
		Escourt		
		Beacon Hill		
		Lowlands		
		Hidcote		
		Mooirivier		
		Rosetta		
30 or *20	115.0 to 128.0	Nottingham Road		
30 or *20	134.5 to 142.0	Balgowan		
		Lidgetton		
		Dargle		
		Lions River		

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Lions River – Bayhead/King's Rest (both directions)

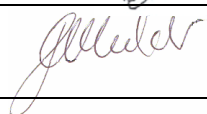
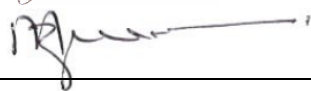
Downgrade speed restrictions		Station	Downgrade speed restrictions	
Speed km/h	km Point		km Point	Speed km/h
30 or *20	146.5 to 150	Lions River Tweedie Merrivale Cedara Boughton	None	None
30 or *20	159.0 to 179			
50 / *35	0 to 2.5	Pietermaritzburg		
		Napier		
		Pentrich		
		Ashburton		
		Umlaas Road		
		Cato Ridge		
		Georgedale		
40 or *25	35 to 62	Hammarisdale		
		Cliffdale		
		Nshongweni		
		Dellville Wood		
		Dassenhoek		
40 or *25	65 to 96	Mariannhill		
		Shallcross		
		Cavendish		
		Mount Vernon		
		Booth		
		Wentworth		
		Bayhead / King's Rest		

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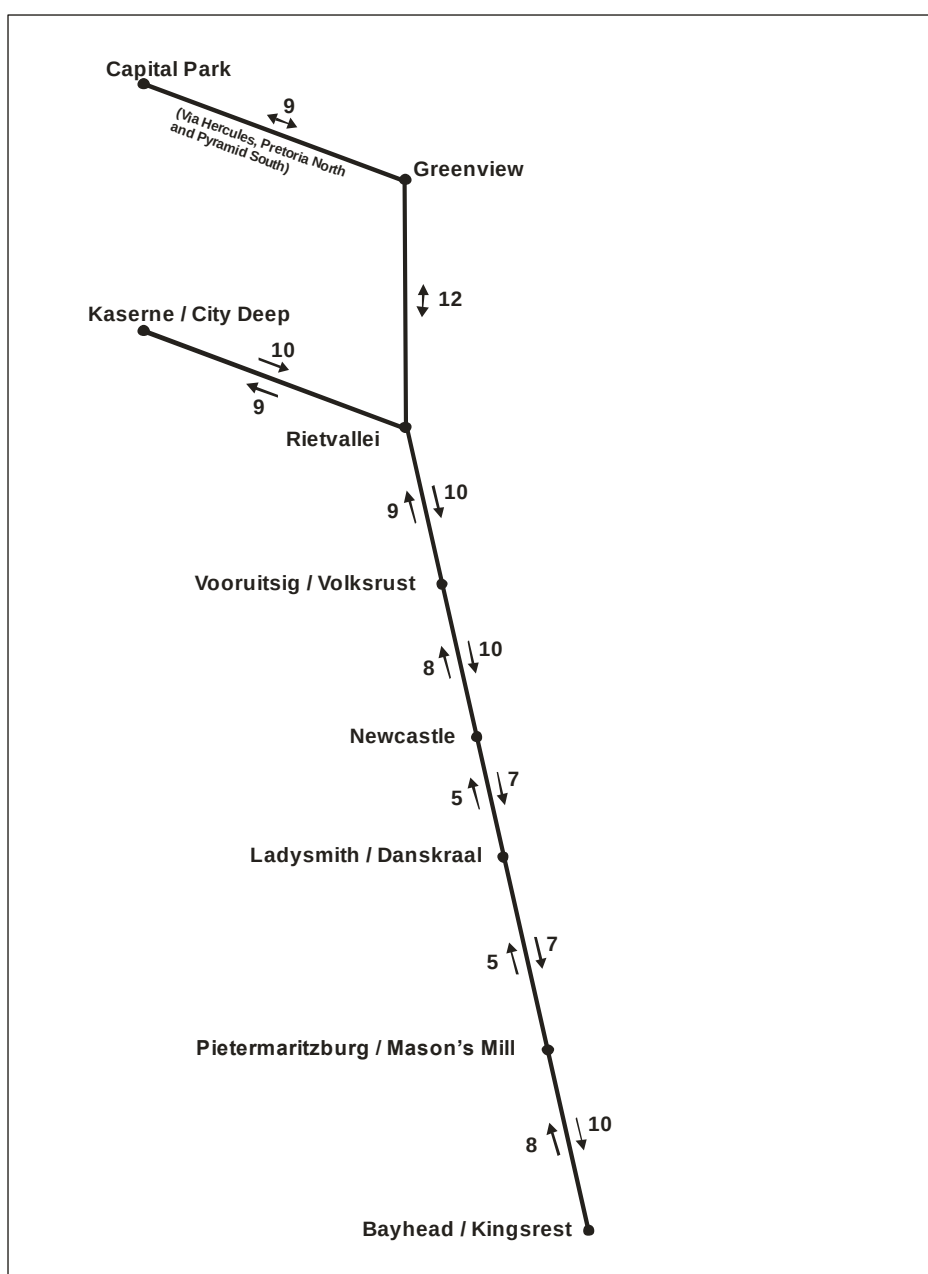
6 LOAD TABLES

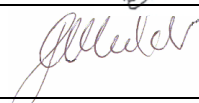
6.1 Traction must be provided in accordance with the Load Tables as contained hereunder,

Gradient Category	=	Actual Gradient
1	=	1 : 30 → 1 : 34
2	=	1 : 35 → 1 : 39
3	=	1 : 40 → 1 : 44
4	=	1 : 45 → 1 : 49
5	=	1 : 50 → 1 : 54
6	=	1 : 55 → 1 : 59
7	=	1 : 60 → 1 : 65
8	=	1 : 66 → 1 : 69
9	=	1 : 70 → 1 : 79
10	=	1 : 80 → 1 : 89
11	=	1 : 90 → 1 : 99
12	=	1 : 100 → 1 : 109
13	=	1 : 110 → 1 : 119
14	=	1 : 120 → 1 : 149
15	=	1 : 150 → 1 : 179
16	=	1 : 180 → 1 : 249
17	=	1 : 250 → 1 : 279
18	=	1 : 280 →

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ROUTE AND GRADIENT CATEGORY INFORMATION

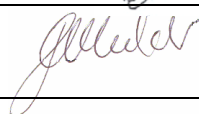
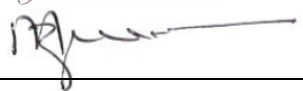


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6.1.1 6E Locomotives

LOAD TABLES FOR CLASS 6E ELECTRIC LOCOMOTIVES AIRBRAKE CONTAINER TRAINS CONSISTING OF A MAXIMUM OF 75 WAGONS

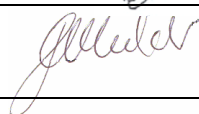
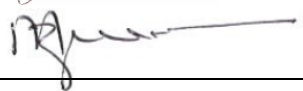
TONNAGE RATINGS BY LINE PER LOCOMOTIVE							
CLASS 6E ELECTRIC LOCOMOTIVES							
GC	GRAD.	CONTAINER TRAINS CLASS : 6E					
		TONNAGE RATINGS BY LINE PER LOCOMOTIVE					
		1	2	3	4	5	6
1	30	340	680	1030	1370	1710	6 x Locomotive Consist Not Authorised
2	35	410	820	1230	1640	2050	
3	40	470	950	1420	1900	2370	
4	45	540	1080	1620	2150	2690	
5	50	600	1200	1810	2410	3010	
6	55	660	1330	1990	2650	3320	
7	60	720	1450	2170	2900	3600	
8	66	800	1590	2390	3180	3600	
9	70	840	1680	2530	3370	3600	
10	80	960	1910	2870	3600		
11	90	1060	2130	3190	3600		
12	100	1170	2340	3510	3600		
13	110	1270	2540	3600			
14	120	1370	2740	3600			
15	150	1650	3290	3600			
16	200	2050	3600				
17	250	2400	3600				
18	280	2580	3600				

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6.1.2 6E1 Locomotives

LOAD TABLES FOR CLASS 6E1 ELECTRIC LOCOMOTIVES AIRBRAKE CONTAINER TRAINS CONSISTING OF A MAXIMUM OF 75 WAGONS

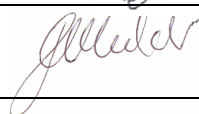
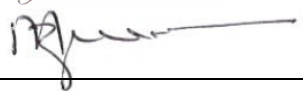
TONNAGE RATINGS BY LINE PER LOCOMOTIVE							
CLASS 6E1 ELECTRIC LOCOMOTIVES							
GC	GRAD.	CONTAINER TRAINS CLASS : 6E1					
		TONNAGE RATINGS BY LINE PER LOCOMOTIVE					
		1	2	3	4	5	6
1	30	410	820	1220	1630	2040	6 x Locomotive Consist Not Authorised
2	35	490	970	1460	1940	2430	
3	40	560	1120	1680	2240	2800	
4	45	630	1270	1900	2540	3170	
5	50	710	1420	2120	2830	3540	
6	55	780	1560	2340	3120	3600	
7	60	850	1700	2550	3390	3600	
8	66	930	1860	2790	3600		
9	70	980	1970	2950	3600		
10	80	1120	2230	3350	3600		
11	90	1240	2480	3600			
12	100	1360	2730	3600			
13	110	1480	2960	3600			
14	120	1590	3190	3600			
15	150	1910	3600				
16	200	2380	3600				
17	250	2780	3600				
18	280	2990	3600				

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6.1.3 10E Locomotives

LOAD TABLES FOR CLASS 10E ELECTRIC LOCOMOTIVES AIRBRAKE CONTAINER TRAINS CONSISTING OF A MAXIMUM OF 75 WAGONS

TONNAGE RATINGS BY LINE PER LOCOMOTIVE CLASS 10E ELECTRIC LOCOMOTIVES							
GC	GRAD.	CONTAINER TRAINS CLASS : 10E					
		TONNAGE RATINGS BY LINE PER LOCOMOTIVE					
		1	2	3	4	5	6
1	30	740	1470	2210	2940	5 & 6 x Locomotive Consists Not Authorised	
2	35	870	1740	2610	3480		
3	40	1000	2000	3000	3600		
4	45	1130	2260	3390	3600		
5	50	1260	2510	3600	3600		
6	55	1380	2760	3600	3600		
7	60	1500	3000	3600	3600		
8	66	1640	3280	3600			
9	70	1740	3470	3600			
10	80	1960	3600	3600			
11	90	2180	3600				
12	100	2390	3600				
13	110	2600	3600				
14	120	2790	3600				
15	150	3340	3600				
16	200	3600	3600				
17	250	3600	3600				
18	280	3600	3600				

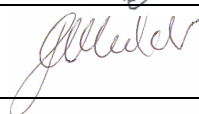
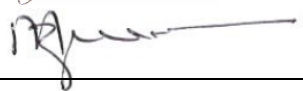
INSTRUCTION FOR THE RUNNING OF AIRBRAKE CONTAINER TRAINS NOT EXCEEDING 75 WAGONS AND/OR 3600 TONS BETWEEN CAPITAL PARK/KASERNE/CITY DEEP AND BAYHEAD/KINGSREST		Document Number : S.GFB/TES/TE/OI 257
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6.1.4 18E Locomotives

LOAD TABLES FOR CLASS 18E ELECTRIC LOCOMOTIVES AIRBRAKE CONTAINER TRAINS CONSISTING OF A MAXIMUM OF 75 WAGONS

TONNAGE RATINGS BY LINE PER LOCOMOTIVE CLASS 18E ELECTRIC LOCOMOTIVES							
GC	GRAD.	CONTAINER TRAINS CLASS : 18E					
		TONNAGE RATINGS BY LINE PER LOCOMOTIVE					
		1	2	3	4	5	6
1	30	430	860	1290	1710	2140	6 x Locomotive Consist Not Authorised
2	35	510	1020	1530	2030	2540	
3	40	590	1170	1760	2350	2940	
4	45	660	1330	1990	2660	3320	
5	50	740	1480	2220	2960	3600	
6	55	810	1630	2440	3260	3600	
7	60	890	1770	2660	3550	3600	
8	66	970	1940	2920	3600		
9	70	1030	2060	3080	3600		
10	80	1160	2330	3490	3600		
11	90	1300	2590	3600			
12	100	1420	2840	3600			
13	110	1540	3090	3600			
14	120	1660	3320	3600			
15	150	1990	3600				
16	200	2480	3600				
17	250	2900	3600				
18	280	3120	3600				

- 6.2** The minimum number of locomotives as indicated in the tables must in all cases be used for traction purposes.
- 6.3** The absolute maximum number of locomotives in a train consist is six.
- 6.4** The load of the train includes all wagons and any dead locomotives. If the mass of any dead locomotives can be accommodated within the allowable load, the number of wagons need not be decreased by the equivalent tonnage.

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- 6.5 If dead locomotives are hauled as part of the load the number thereof, may not exceed the number of active locomotives in the consist
- 6.6 The electric brakes of active locomotives in a train consist must be engaged to a maximum of four. The exception being classes 6E/6E1, where the electric brakes of a maximum of three locomotives in the consist must be engaged.
- 6.7 Prior to the departure of the train it must be ensured that the number of locomotives as provided for traction purposes are acceptable and prove adequate for the charging of the train brake system. If difficulty is experienced in the charging of the train brake system an examination for potential leaks should be performed. Provided no leaks are identified either additional locomotive/s must be provided to facilitate the charging of the train brake system, or alternatively an appropriate number of wagons should be removed from the train.

7 RUNNING TIMES

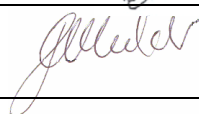
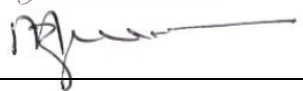
- 7.1 Taking cognisance of paragraph 4 (above), trains should be operated utilising running times as contained in the Master Train Schedule (MTS) as a guideline.

8 BRAKE PIPE PRESSURE SETTINGS

- 8.1 Where the average axle mass load of the train exceeds 15 tons per axle : 550 kPa
- 8.1.1 On trains where the average axle mass load of the train exceeds 15 tons per axle no empty container wagons (wagons devoid of any containers) are permitted on the train.
- 8.2 Where the average axle mass load of the train is 15 tons per axle or less : 500 kPa

9 WAGON COMPILATION RULES

- 9.1 The maximum train lengths are specified in Paragraph 11 of this document.
- 9.2 By virtue of direct reference to document RSE/TE/LPR/0040 (Revision 2) dated 26 July 2016 and titled 'Loading Specification for Securing and Transporting of Containers' as issued by Dr.

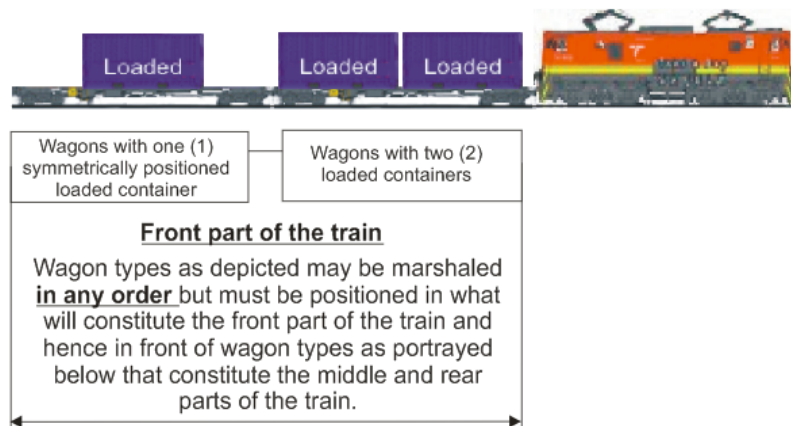
INSTRUCTION FOR THE RUNNING OF AIRBRAKE CONTAINER TRAINS NOT EXCEEDING 75 WAGONS AND/OR 3600 TONS BETWEEN CAPITAL PARK/KASERNE/CITY DEEP AND BAYHEAD/KINGSREST		Document Number : S.GFB/TES/TE/OI 257
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R. Frohling of the Technology Management (Mechanical Technology) department the following general train compilation rules will apply:

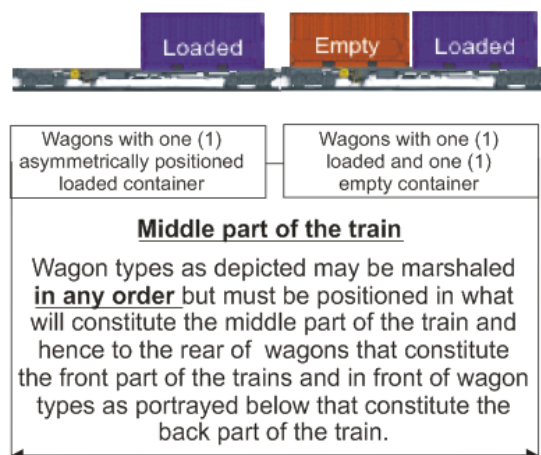
The train should be marshaled in the following order –

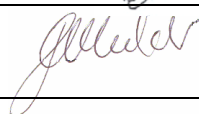
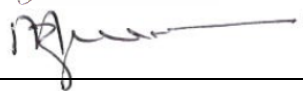
Trains with an average axle mass load of 15 tons per axle or less

Front part of the train

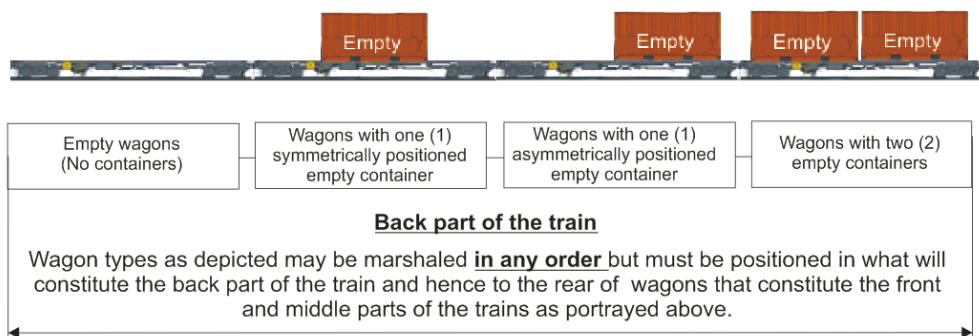


Middle part of the train



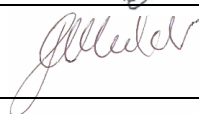
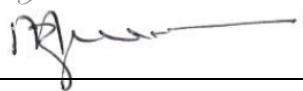
INSTRUCTION FOR THE RUNNING OF AIRBRAKE CONTAINER TRAINS NOT EXCEEDING 75 WAGONS AND/OR 3600 TONS BETWEEN CAPITAL PARK/KASERNE/CITY DEEP AND BAYHEAD/KINGSREST		Document Number : S.GFB/TES/TE/OI 257
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Back part of the train



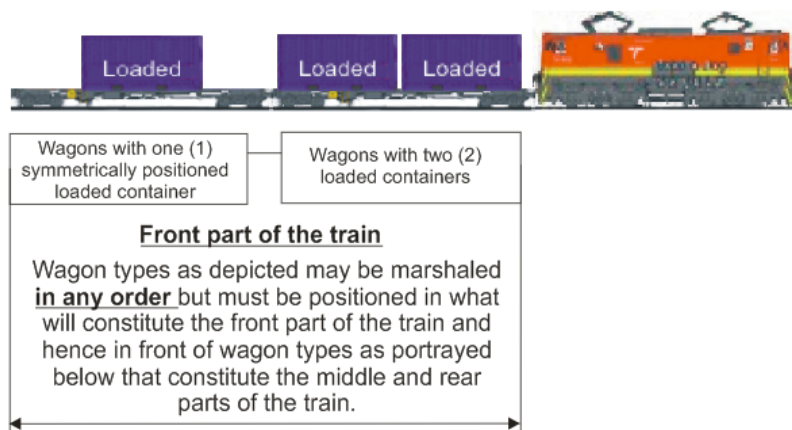
The wagon compilation rules as depicted above are graphically portrayed below to promote a better understanding and to aid simplification however it must be noted that wagon types displayed in the particular sections (front, middle and rear) may be marshaled in any order relevant to the designated segment only.



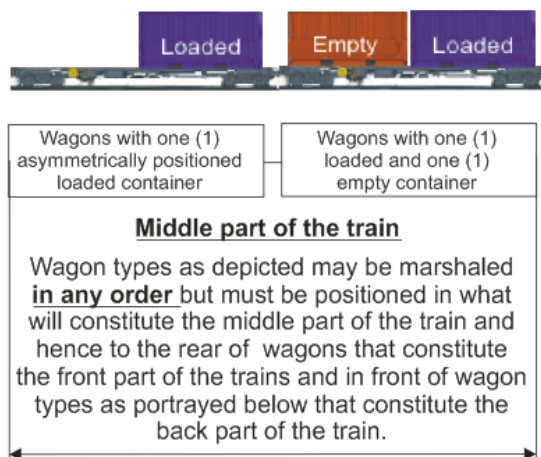
INSTRUCTION FOR THE RUNNING OF AIRBRAKE CONTAINER TRAINS NOT EXCEEDING 75 WAGONS AND/OR 3600 TONS BETWEEN CAPITAL PARK/KASERNE/CITY DEEP AND BAYHEAD/KINGSREST		Document Number : S.GFB/TES/TE/OI 257
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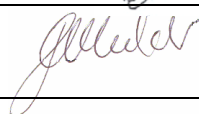
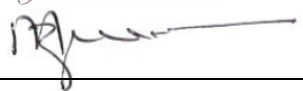
Trains with an average axle mass load exceeding 15 tons per axle

Front part of the train

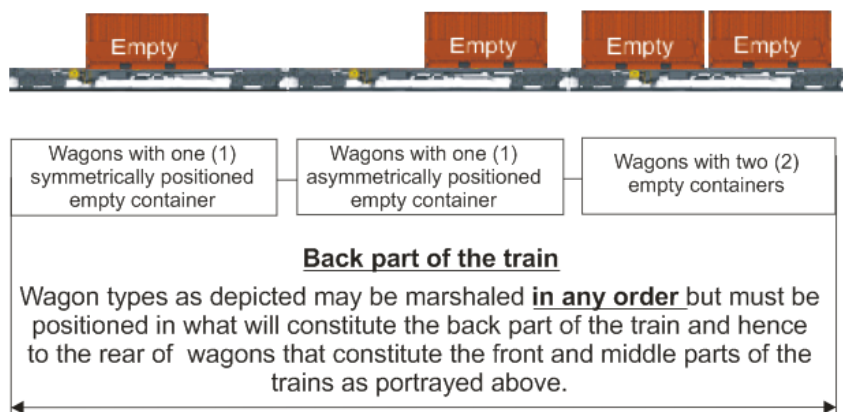


Middle part of the train



INSTRUCTION FOR THE RUNNING OF AIRBRAKE CONTAINER TRAINS NOT EXCEEDING 75 WAGONS AND/OR 3600 TONS BETWEEN CAPITAL PARK/KASERNE/CITY DEEP AND BAYHEAD/KINGSREST		Document Number : S.GFB/TES/TE/OI 257
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Back part of the train

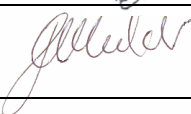
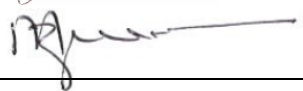


The wagon compilation rules as depicted above are graphically portrayed below to promote a better understanding and to aid simplification however it must be noted that wagon types displayed in the particular sections (front, middle and rear) may be marshaled in any order relevant to the designated segment only.



Refer to sub-clause 8.1.1 above where it is stated that on trains where the average axle mass load of the train exceeds 15 tons per axle no empty container wagons (wagons devoid of any containers) are permitted on the train

- 9.3** If the number of locomotives as prescribed in the load tables exceeds four 6E/6E1/18E locomotives or three Class 10E electric locomotives respectively, then no wagons fitted with Saskop couplers are permissible within the first ten wagons behind the locomotives.
- 9.4** If wagons containing dangerous goods are coupled to the train, separation wagons must be provided in accordance with section 1 of the General Appendix (Part I).

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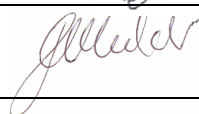
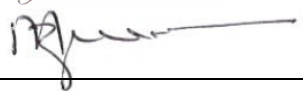
10 WAGON TYPES

- 10.1** Only approved container wagons equipped with airbrakes and roller bearings may be hauled on these trains.
- 10.2** Relevant to medium-to-heavy container trains only approved wagons designed for such loads and equipped with airbrakes and roller bearings may be hauled on such trains.

11 TRAIN LENGTH AND ROUTE

- 11.1** The maximum length of the trains authorized in this instruction is 75 wagons or 3600 tons, dependent on whichever comes first. (both directions)
- 11.2** To emphasize the permissible train length allowable it must be noted that the absolute maximum number of wagons is restricted to seventy five (75) and the maximum allowable tonnage is restricted to 3600 tons **whichever comes first**. Therefore it can be established that to have a trainload of 75 wagons that is 3600 tons will result in the average axle mass load not exceeding 12 tons per axle ($12\text{t/a} \times 4 \text{ axles} \times 75 \text{ wagons} = 3600 \text{ tons}$).
- 11.3** The approved route for container trains to and from Capital Park will be via Greenview en-route to the Natal Mainline.



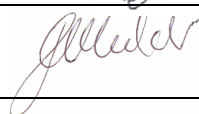
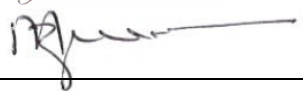
INSTRUCTION FOR THE RUNNING OF AIRBRAKE CONTAINER TRAINS NOT EXCEEDING 75 WAGONS AND/OR 3600 TONS BETWEEN CAPITAL PARK/KASERNE/CITY DEEP AND BAYHEAD/KINGSREST		Document Number : S.GFB/TES/TE/OI 257
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12 DIFFICULT SIGNALS

12.1 The following signals are difficult signals for all loaded airbrake container trains.

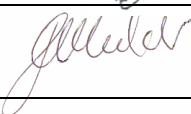
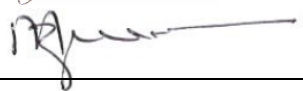
1. Kaserne / City Deep / Capital Park to Bayhead / Kings Rest direction

Kaydale – Vooruitsig	Vooruitsig - Ladysmith	Ladysmith – Bayhead / Kings Rest
KXL140/142 BOR640/642 BOR840/842 FI140/142 FI240/242 KMI140/142 KMI240/242 PRD140/142 PRD240/242 PD140/142 PD240/242	MIX860/858 MIX760/758 DR860/858 WBK860/858 WSN860/858 WSN760/758 ELL460/458	UB160/158 CV860/858 FE360/358 FE260/258 FE160/158 ENF860/858 ENF660/658 ENF160/158 EN660/658 BEC660/658 BEC860/858 BCH860/858 BCH160/158 LOW860/858 LOW660/658 LOW160/158 HH960/958 RMR960/958 RA160/158

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2. Bayhead / Kings Rest to Kaserne / City Deep / Capital Park direction

Bayhead / Kings Rest - Ladysmith	Ladysmith – Kaydale
SCS38/40	UTK138/140
DAS938/940	WLS738/740
DDW38/40	WLS838/840
DLE38/40	GJN1338/1340
GDL38/40	SLL840/842
GDL138/140	IGO140/142
COR138/140	IGO240/242
COR238/240	IGO840/842
PMB838/840	GHT840/842
PMB938/940	MPP140/142
BHN138/140	MPP840/842
BHN238/240	LGK140/142
BHN338/340	LGK840/842
BHN838/840	PMD760/762
BSE138/140	SNA260/262
BSE638/640	SNA280/282
CDA38/40	SNA162
CDA138/140	SYU762/760
CDA338/340	SYU860/862
MRV838/840	
TD138/140	
DRE138/140	
DRE338/340	
LN138/140	
LN338/340	
LN938/940	
BG138/140	
BG338/340	
BG738/740	
BG838/840	
BG938/940	
HLV138/140	
HLV638/640	
NR138/140	
NR238/240	
NR338/340	
PI238/240	

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13 PERSONNEL ARRANGEMENTS

- 13.1 The relevant Regions must perform all necessary train/train personnel arrangements.

14 OVERLOADING

The following clauses are highlighted with reference to document BBD 5249 and titled '*Alarm Limits for the Measurement of Weighbridges, Wheel Impact Monitors and Skew Bogie Detectors*'.

14.1 Overloading.

14.1.1 Relevant to airbrake trains as authorized by this document the maximum allowable overload permitted for the train to continue is 'One (1) ton X the Number of Wagons on the Train'. Thereafter a sufficient number of wagons must be removed from the train to ensure that the train load is kept within the specified limits.

14.1.2 The train must be stopped where the gross wagon mass of any wagon is identified as being 7% or higher for the specific wagon. Any such identified wagon should be removed from the train accordingly.

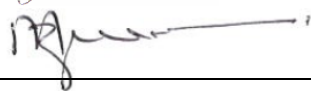
15 SPECIAL ARRANGEMENTS

- 15.1 The relevant Train Driver/s responsible for the handling of such trains must be supplied with a copy of this Authority.

- 15.2 Hauler Movements between Bayhead and Kings Rest.

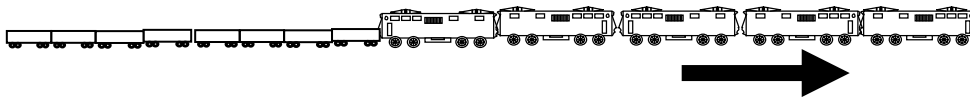
15.2.1 Hauler movements are authorized between Bayhead and Kings Rest to facilitate the offloading of container trains. The wagon compilation rules as specified above (Point 8) will not apply to these train movements. The normal prescribed brake tests will be applicable to all such train movements and an absolute maximum speed of 20 km/h may not be exceeded.

15.2.2 In the event of such train movements as authorized above Train Drivers must exercise caution when making use of the independent brake facility, this to alleviate any undue in-train action and/or forces.

INSTRUCTION FOR THE RUNNING OF AIRBRAKE CONTAINER TRAINS NOT EXCEEDING 75 WAGONS AND/OR 3600 TONS BETWEEN CAPITAL PARK/KASERNE/CITY DEEP AND BAYHEAD/KINGSREST		Document Number : S.GFB/TES/TE/OI 257
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1st Technical Approving Officer	A.R. Boy : 	Issue date: 2016-08-15
2nd Technical Approving Officer	J.M. Mulder : 	Page Number : 26 of 26
Approving Safety Officer:	T. Zwane : 	

- 15.3 A copy of this Technical Train Operating Instruction has also been made available on the ProjectWise platform.





16 Anaconda Train Handling Guide, From Union To Booth- Revision 1

5 x 18E + 75 x Container Wagon Train (or 3600ton load)

Only 4x 18E in Rheostat

City Deep To Union

- Control with traction out of City Deep up to Refinery, before km. 1
- Around km. 1 take rheostat to control speed
- Join line around Elsburg at 30km/h

Union To Vooruitsig

- Through Union station, control with rheostat braking at train speed of 30km/h where relevant
- Control train speed below 50km/h once out of Union,
- At km 4, just before Rooikop, train speed must be 60km/h to confirm signals and cross over to mainline
- If on Suidrand line control train speed below 30km/h to cross over diamond crossing back to Natal line at km. 5/11
- Once in Rooikop, take traction and accelerate train, controlling speed at 60km/h
- Around Mapleton (km. 14.5) decrease traction and take rheostat brake and allow speed to increase under control below 60km/h
- Make train brake application if or as necessary, to control train speed at 60km/h
- Looking at accelerometer around km. 17, release all brakes (if applied) and allow train to swing to 80km/h
- Take traction gradually once speed drops and control speed at 60km/h
- After Glenroy, around km. 23/9, change over to idle or light rheostat brakes to control train at 60km/h, through Driemanskap (km. 27.5)
- Take light traction around km.33.5, before Kaydale, to maintain speed and switch back to rheostat when needed

- After km. 39.5, before Heidelberg, take traction as to control speed at 60km/h
- After km. 41 at the exit of Heidelberg control speed at 60km/h with traction through Kraal (km. 49)
- Before km. 55, as you tip, introduce rheostat brakes to control train speed
- *After Spruytsrus, from km. 58, speed must be controlled at 50km/h, to adhere to down gradient speed restriction, up to km. 61.5*
- After km. 61.5, looking at accelerometer, allow speed to increase to 80km/h,
- Take traction as needed and control speed at 60km/h towards and through Balfour
- Around km. 78.5 decrease traction and switch over to rheostat brakes (if necessary) to control train speed at 60km/h
- Around km. 94 take traction to control speed at 60km/h
- At about km. 97 change to rheostat brakes to control train speed at 50km/h to adhere to down gradient speed restriction up to km. 99
- Go into idle around km. 101 and allow train to swing to 80km/h
- Take traction gradually around km. 102.5, to decrease run-in and control train speed at 60km/h.
- From km. 108 look at accelerometer and switch over to rheostat to control speed at 60km/h
- *Km. 116.5, Energy Saving Swing, thus allow train to swing to 80km/h*
- Around km. 118 introduce traction to control speed at 60km/h, until km. 132, allowing train to idle through Holmdene
- After km. 132.5 take rheostat brakes to control speed at 60km/h
- *Km. 139.5 Energy Saving Swing, thus allow train to swing to 80km/h*
- At km. 141 switch over to traction to maintain speed at 60km/h and decrease run-in
- From km. 144 looking at accelerometer take rheostat to control speed below 60km/h
- Speed must be controlled at 60km/h in and out of Standerton
- From km 148.5 speed must be controlled at 60km/h, to adhere to curve speed restriction
- From km. 149.5 introduce traction to maintain speed at 60km/h
- *Km. 139.5 Energy Saving Swing, thus allow train to swing to 80km/h*

- Around km.159 take traction to control speed at 60km/h through Kromdraai, Kanonheuwel, and Platrand
- *Km. 180 Energy Saving Swing, allow speed to increase*
- Km. 181 increase traction and control speed up to Rusthof, km. 185
- After Rusthof, km. 186, close traction and either take rheostat or light brake application to control train speed at 80km/h???
- Around km.188 take traction to control speed at 60km/h up to Perdekop, km. 195
- *Km. 194 Energy Saving Swing, thus allow train to swing to 80km/h*
- Increase traction gradually to maintain speed at 60km/h
- At Elandspoort, around km. 200.5 change over to rheostat to control train speed, at 50km/h to adhere to down gradient speed restriction until km. 203
- *From km 204.5 speed must be controlled at 50km/h, to adhere to down gradient speed restriction, up to km 208.5*
- Train brakes must be applied if or as needed to control train speed
- Before Beechwick km. 208.5, *(once difficult signal was confirmed)* allow speed to increase and control speed at 60km/h
- *Km. 215 Energy Saving Swing, thus allow train to swing to 80km/h*
- Around km. 216.5 take traction to control speed at 60km/h through Sandspruit
- Around km. 220.5 decrease traction and change over to rheostat braking to control speed at 60km/h
- At km. 223 introduce traction to decrease run-in and to control speed at 60km/h up to Vooruitsig
- *Enter Vooruitsig at 20km/h*

Vooruitsig To Ladysmith

- Once through Vooruitsig around km. 0.5 take rheostat brakes to control speed
- *From km. 0.5 speed must be controlled at 40km/h, to adhere to down gradient speed restriction, up to km. 4.5*
- Make train brake application if or as needed
- Around km. 5.5 switch to traction to control speed at 60km/h

- *From km. 9 speed must be controlled at 40km/h with rheo, to adhere to down gradient speed restriction, up to km. 33.5*
- Make train brake application if or as needed
- *From km. 35.5 speed must be controlled at 40km/h, to adhere to down gradient speed restriction, up to km. 40*
- Around km. 40 release train brakes and allow speed to increase to and control speed at 60km/h with rheostat through Clontarf
- *Km. 48.5 Energy Saving Swing, thus allow train to swing to 70km/h*
- Watch accelerometer and around km. 49.5 introduce traction as required to maintain speed at 60km/h
- Before km. 54.5 decrease traction and switch to rheostat brakes to control train speed at 40km/h until km. 56
- Around km. 59.5 take traction to maintain speed at 60km/h through New Castle
- Around km. 67.5, close to Rooikop, tip carefully and take rheostat brake to control speed
- *Allow speed to drop to 40km/h at km. 68 to adhere to speed restriction until km. 70.5*
- *Km. 70.5 Energy Saving Swing, thus allow speed to increase to 80km/h*
- Before km. 71.5 take traction to control speed at 60km/h and minimize run-in
- Around km. 73 switch to rheostat to control speed at 60km/h
- Around km. 75 take traction and control speed at 60km/h, through Ngagane and Ballengeich
- In traction maintain speed at 60km/h going through Milnerdale and Danhauser
- *Km. 96 Energy Saving Swing, thus allow train to swing to 80km/h*
- *Around km. 97 increase traction to maintain speed at 60km/h*
- *From km. 102.5 speed must be controlled at 50km/h with rheostat, to adhere to down gradient speed restriction, up to km. 107*
- Make train brake applications if and as needed to control speed
- Allow speed to increase and control speed at 60km/h through Hattingspruit
- Around km. 110 take traction to control train speed at 60km/h
- After km. 117 introduce rheostat to control train speed

- After km. 118 enter and pass through container wagon inspection site at 5km/h with light traction
- Enter and exit Glencoe at 30km/h, around km. 120.5
- Once out of Glencoe, allow speed to increase to 40km/h
- *From km. 121 speed must be controlled at 30km/h, to adhere to down gradient speed restriction, up to km. 131*
- *From km 131 speed must be controlled at 40 km/h, to adhere to down gradient speed restriction, up to km 136*
- Km. 136 Energy Saving Swing, thus allow speed to increase to 80km/h
- At km. 137 take traction to control speed at 60km/h toward Wasbank
- *Around, at Wasbank (km. 140), apply and release a minimum application to prevent run-in and keep light traction to assist train to reach 80 km/h*
- *Keep traction to maintain speed at 60km/h*
- Around km. 148, Wesselsnek, decrease traction and take rheostat as the train tips to control speed at 40km/h
- After Maduna, around km. 156 introduce traction to maintain speed at 60km/h
- At km. 160.5 enter Elandslaagte at 40km/h (apply train brakes when necessary to reduce speed)
- Exit Elandslaagte, at speed restriction of 50km/h and then allow speed to increase to and control speed at 60km/h
- Before km. 164 decrease traction and take rheostat to control train speed at 60km/h
- Around km. 169.5 take traction to maintain speed at 60km/h
- Around km. 173.5 switch to rheostat brakes to control speed
- Around km. 175.5 take traction to maintain speed and idle once tipped
- *From km. 177.5 speed must be controlled with rheostat at 40km/h, to adhere to down gradient speed restriction, up to km. 181*
- After km. 181 enter Danskraal North at 30km/h
- Enter Ladysmith with train speed of 30km/h

Ladysmith To Pietermaritzburg

- Exit Ladysmith at 30km/h and allow speed to increase to 60km/h
- Use rheostat to control speed
- Km. 3 Energy Saving Swing, thus allow speed to increase to 70km/h
- From km 6 speed must be controlled at 60km/h, to adhere to curve speed restriction,
- At km 7 speed must be controlled at 50km/h, to adhere to curve speed restriction,
- After km. 8 take traction to control speed at 60km/h
- Km. 12.5 Energy Saving Swing, thus allow train to swing, increasing speed to 70km/h going towards Pieters
- At km 14 speed must be controlled at 70km/h, to adhere to curve speed restriction,
- *From km 15 speed must be controlled with rheostat braking at 30km/h, to adhere to down gradient speed restriction, up to km 18*
- Train brakes must be used when and as required.
- Km. 18.5, allow train to swing, speed to increase to 70km/h,
- Introduce traction gradually as needed to maintain speed at 60km/h,
- After km 23 speed must be controlled at 50km/h, to adhere to curve speed restriction, around Colenso
- When train cleared the curve, increase traction to control speed at 60km/h through Chieveley and further
- Around km. 38.5 (while tipping), decrease traction and introduce rheostat braking to control speed at 60km/h
- Around km. 42 keep speed around 50km/h to confirm difficult signal before Frere
- Once the signal is confirmed, speed can be increased to 60km/h to adhere to curve speed restrictions after km 43, until the train cleared the series of curves
- Around km. 43.5 take traction to maintain speed at 60km/h through Heavitree
- Ensure speed is below 60km/h through curves at km 50.5 and after km 52.5, to adhere to curve speed restrictions
- Once through the curves and at Ennersdale (after km. 54), increase speed and keep below 60km/h
- After km. 57 take rheostat to control speed at 60km/h
- Before km. 60 control speed at 60km/h through tunnel
- Through Estcourt, at km. 64.5, speed must be controlled at 60km/h,

- From km. 65 speed close to idle and allow speed to increase to swing train at 70km/h, while adhering to curve speed restrictions
- Before km. 67 change over to traction to maintain and control speed at 60km/h through Beacon Hill and Lowlands
- After km. 83 control speed through tunnel at 50km/h
- At km. 88.5, introduce rheostat and control speed at 50km/h to adhere to downgradient speed restriction up to km. 93
- Before Mooi River, after km 96, speed must be controlled at 60km/h, to adhere to curve speed restrictions
- Enter and exit Mooiriver at 50 km/h
- Km. 98 decrease rheostat and introduce traction to increase and control speed at 60km/h
- Around km. 103.5 decrease traction and switch to rheostat to control speed (apply a brief minimum application if needed)
- Around Rosetta, after km. 108, speed must be controlled at 60km/h, to adhere to curve speed restriction
- After km. 108, looking at accelerometer take traction to control speed at 60km/h
- After km. 113.5 before Nottingham Rd, speed must be controlled at 60km/h, to adhere to curve speed restriction,
- After Nottingham Rd, km 114.5, speed must be controlled at 60km/h, to adhere to curve speed restriction,
- *Before km. 115, introduce rheostat to reduce and control speed at 30km/h, until km.128 to adhere to down gradient speed restriction*
- Looking at accelerometer apply train brake when and as required to maintain 50km/h
- Around km. 128, release train brakes and control speed at 50km/h with only rheostat to negotiate level and adhere to curve speed restriction,
- Introduce traction when required to maintain the speed at 50km/h
- *From km. 134, introduce rheostat and apply train brake to control the speed at 30km/h until km. 142 to adhere to down gradient speed restriction,*
- After km. 142.5, control speed at 50km/h to adhere to curve speed restrictions
- Control speed over level crossing at 40km/h

- Around km 143, decrease rheo and introduce traction to maintain speed at 50km/h to adhere to curve speed restrictions,
- After km. 146.5, introduce rheostat to control speed at 30km/h to adhere to down gradient speed restriction through Tweedie to km. 149.5
- Apply train brakes around km. 148 if or when necessary to control speed
- At km. 149.5, release train brakes and allow train to swing at 70km/h
- Looking at accelerometer, take traction to maintain speed, but ensure train is at 60km/h after km. 152.5 to adhere to curve speed restriction
- Around km. 154 take rheostat braking and ensure speed is controlled at 50km/h, to adhere to curve speed restriction before Merrivale,
- After km. 155.5 take traction to maintain speed and ensure curve speed restriction of 60km/h is adhered to
- Around km. 157.5 take rheostat and when needed apply train brakes to control train speed
- *From km. 159, speed must be controlled at 30km/h to adhere to down gradient speed restriction until km. 179, through Cedara, Boughton and Rushbrook*
- Enter Pietermaritzburg at 30km/h

Pietermaritzburg To Booth

- Exit Pietermaritzburg with rheostat brakes at 30km/h
- After km 1.5, speed must be controlled at 40km/h, to adhere to curve speed restrictions,
- Around km. 3 introduce traction to maintain speed
- After km. 3.5 increase speed while adhering to passing curve speed restrictions of 50km/h and 60km/h around km. 3
- Around km. 10.5 inside the tunnel decrease traction and introduce rheostat to control speed at 60km/h
- At km. 12 decrease rheo and allow train to swing at 80km/h,
- Take traction and control speed at 60km/h, until km. 26

- At km. 26, decrease traction and introduce rheostat to control speed at 60km/h, to adhere to curve speed restriction,
- At km. 29, speed must be controlled at 50km/h, to adhere to curve speed restrictions,
- At km. 31 allow speed to increase to 60 km/h to adhere to curve speed restrictions
- Prepare to exit Cato Ridge at 30km/h for cross over, can use train brake to reduce the speed
- *From km. 35, speed must be controlled at 40km/h to adhere to down gradient speed restriction until km. 38,*
- *From km. 38, speed must be controlled at 50km/h to adhere to down gradient speed restriction until km. 51, unless lower curve speed restrictions are imposed*
- Use train brakes when and as required to control the speed
- After km 51, speed must be controlled at 40km/h, to adhere to curve speed restriction,
- From km. 53 speed may increase slightly to 45km/h around km. 54 as train brakes are released
- After km 54.5 up to km. 60, speed must be controlled at 40km/h, to adhere to curve speed restrictions, (take traction if needed through Cliffdale)
- Around km. 56 take rheostat brakes to control train speed
- *From km. 60, a down gradient speed restriction of 50km/h are imposed until km. 62,*
- After km. 60, allow speed to increase to 50km/h until km. 62 to adhere to curve speed restrictions
- After km. 62, increase speed to 60km/h, take traction around km. 63
- *From km. 65, a down gradient speed restriction of 50km/h are imposed until km. 96, but lower curve speed restrictions are specified*
- Around km. 65 introduce rheostat and apply train brakes when and as required to control speed
- From km. 66 until km. 78 speed must be controlled at 40km/h to adhere to curve speed restrictions,
- From km. 78 until km. 86 speed can be controlled at 50km/h to adhere to curve speed restrictions,
- From km. 86 until km. 88.5 speed must be controlled at 40km/h to adhere to curve speed restrictions,

- From km. 88.5 until km. 91 speed can be controlled at 50km/h to adhere to curve speed restrictions,
- From km. 91 until km. 93 speed must be controlled at 40km/h to adhere to curve speed restrictions,
- After km. 93.5 speed must be controlled at 50km/h to adhere to curve speed restriction,
- Use Rheostat to control speed as required through the curves until you enter Booth
- Enter Booth at 30km/h

Booth To Kingsrest

- Control train speed with rheostat at 30km/h
- Into Kingsrest at 8km/h



Anaconda Train Handling Guide, From Booth To Union - Revision 1

5 x 18E + 75 x Container Wagon Train (or 3600ton load)

Only 4x 18E in Rheostat

Kingsrest To Booth

- Depart from Kingsrest and control speed at 30km/h, mainly with traction
- Use brakes only after the flyover

Booth To Pietermaritzburg

- Continue through from Booth at 30km/h with traction
- Around km. 96 increase traction and control train speed at 50km/h to adhere to curve speed restrictions
- After km. 92, control train speed at 40km/h to adhere to curve speed restriction
- Around km. 90 increase speed to 50km/h to adhere to curve speed restriction
- Before km. 87, control train speed at 40km/h to adhere to curve speed restriction
- From km. 85, control train speed at 50km/h to adhere to curve speed restrictions, up to km. 77
- After km. 77 control train at 40km/h to adhere to curve speed restrictions, until after km. 69.5
- After km. 69.5, control train at 50km/h to adhere to curve speed restrictions, until after km. 67
- After km. 67, speed must be controlled at 40km/h, to adhere to curve speed restrictions
- From km. 65 control train speed with traction at 60km/h
- After km. 61.5, speed must be controlled at 50km/h, to adhere to curve speed restrictions
- At km. 59, speed must be controlled at 40km/h, to adhere to curve speed restrictions, until after km. 53.5 with traction

- After km. 53.5, speed can be controlled at 50km/h, to adhere to curve speed restrictions,
- After km. 52, speed must be controlled at 40km/h, to adhere to curve speed restriction,
- Before km. 50, speed must be controlled at 50km/h, to adhere to curve speed restrictions,
- Around km. 47.5, speed can be increased to 60km/h, to adhere to curve speed restrictions,
- Before km. 43.5, speed must be controlled at 50km/h, to adhere to curve speed restrictions, until before km. 34
- Prepare at km. 35.5 to enter Cato Ridge at 30km/h for cross over, controlling train with traction
- Close off traction to M1 and allow train to increase speed
- Once out of station, around km. 33, speed must be controlled at 60km/h, to adhere to curve speed restrictions,
- Before km. 30, around Camperdown, speed must be controlled at 50km/h, to adhere to curve speed restrictions,
- Around km. 28, speed can be increased and controlled at 60km/h, to adhere to curve speed restriction,
- Around km. 25 take rheostat braking and allow speed to increase and control speed below 60km/h
- Around km. 14 allow speed to increase to swing at 80km/h
- Around km. 12.5 take traction to maintain speed
- Around km. 10.5 introduce rheostat to control train speed below 60km/h
- At km. 6, speed must be controlled at 60km/h, to adhere to curve speed restriction,
- If on line 2, speed out of Pentrich must be controlled at 30km/h for the diamond crossing
- Around km. 3.5 take traction to control speed
- After km. 3.5, speed must be controlled at 60km/h, to adhere to curve speed restrictions,
- After km. 3, before Mason's Mill, speed must be controlled at 50km/h, to adhere to curve speed restrictions,

- If on line 1 at Masons Mill points, speed must be limited to 30km/h for diamond crossing
- After km. 2/19, speed must be controlled at 40km/h, to adhere to curve speed restrictions,
- Enter and exit Pietermaritzburg at 20km/h (for the double yellow signal) permanent speed restriction through PMBurg

Pietermaritzburg To Ladysmith

- Allow speed to increase from 20km/h with traction
- Once out of the station, around km. 178, speed can be increased and controlled at 50km/h, to adhere to curve speed restrictions, up to km 173.5
- After km. 173.5, speed must be controlled at 40km/h, to adhere to curve speed restrictions, up to km 169.5
- Increase traction and allow train to reach balancing speed on the up gradient
- Around km. 167.5, speed must be controlled at 60km/h, to adhere to curve speed restriction
- Continue with traction through the Cedara tunnel (Tunnel speed 60 km/h)
- Around km. 156.5 take rheostat to control train speed at 60km/h, to adhere to curve speed restriction,
- Before km. 154.5, around Merrivale, speed must be controlled at 50km/h, to adhere to curve speed restriction (take traction when needed to maintain speed)
- Before km. 153, around Quail, speed must be controlled at 60km/h, to adhere to curve speed restriction,
- Around km. 152.5 take rheostat braking to control speed,
- Km. 151 allow train speed to increase and swing to 70km/h
- Around km. 150 take traction to control speed
- After km. 147.5, around Tweedie, speed must be controlled at 60km/h, to adhere to curve speed restriction,
- Around km. 146 take rheostat brakes to control speed at 50km/h, to adhere to curve speed restrictions, up to km. 139.5

- Around km. 142.5 take traction to control speed at 50km/h
- Before km. 138 speed must be controlled at 50km/h, to adhere to curve speed restriction,
- Before km. 135.5 speed must be controlled at 50km/h, to adhere to curve speed restriction,
- Before km. 133.5 speed must be controlled at 60km/h, to adhere to curve speed restrictions,
- After km. 132 speed must be controlled at 50km/h, to adhere to curve speed restrictions, up to km. 125.5,
- Around km. 125.5 speed can be increased and controlled at 60km/h to adhere to curve speed restrictions
- Before km. 122 speed must be controlled at 50km/h, to adhere to curve speed restrictions, up to km. 116.5
- After km. 115 speed must be controlled at 60km/h, to adhere to curve speed restriction,
- Around km. 113.5 take rheostat brakes and control speed at 60km/h
- Around km. 108.5 speed must be controlled at 60km/h, to adhere to curve speed restriction,
- Around km. 107 take traction to maintain speed at 60km/h
- Around km. 103.5 speed must be controlled at 50km/h, to adhere to speed restriction through tunnel
- *Around km. 103 speed must be controlled at 40km/h, to adhere to down gradient speed restriction, up to km. 99*
- Around km. 103 take rheostat and apply train brakes when and as needed to control speed
- Before km. 97.5 speed must be controlled at 60km/h, to adhere to curve speed restrictions,
- Enter and exit Mooiriver at 50km/h
- Around km. 97 take traction to maintain speed
- From km. 95 increase speed to 60km/h
- *From km. 88 change over to rheostat and control speed at 30km/h for Line nr. 2 and 40km/h for Line nr. 1, to adhere to down gradient speed restriction, up to km 81*

- Apply train brakes when and as needed to control speed
- *From km. 80 speed must be controlled at 40km/h, to adhere to down gradient speed restriction, up to km. 67 and release train brakes*
- *Km. 67 Energy Saving Swing, thus allow train to swing to 70km/h*
- *Around km. 66 take traction to maintain at 60km/h*
- After km. 66 speed must be controlled below 70km/h, to adhere to curve speed restrictions,
- Km. 60, control speed at 60km/h through tunnel
- Around km. 56 take rheostat brakes to control train speed
- Before km. 53 speed must be controlled at 60km/h, to adhere to curve speed restriction,
- *From km. 51 speed must be controlled at 40km/h, to adhere to down gradient speed restriction, up to km. 44*
- Apply train brakes when and as needed to control speed
- After km. 44 speed must be controlled at 60km/h, to adhere to curve speed restrictions,
- Around km. 43 release train brakes
- After km. 43, around Frere, speed must be controlled at 60km/h, to adhere to speed restriction over steel bridge,
- Around km. 42.5 take traction and control speed
- Km. 38, tip slow enough to control train speed below 50km/h
- From km. 37.5 take rheostat and if apply train brakes when needed to control speed
- Release train brakes after km. 36
- *From km. 31 speed must be controlled at 40km/h, to adhere to down gradient speed restriction, up to km. 26*
- Around km. 31 apply train brakes when and as needed to control speed
- Around km. 25 release train brakes and control train with rheostat only, allow the train speed to increase to 50 km/h
- Into and out of Colenso, around km. 23.5, take traction and control speed at 50km/h, to adhere to speed restriction,
- At km. 22 increase speed and control at 60km/h

- After km. 14.5, speed must be controlled at 70km/h, to adhere to curve speed restriction,
- Km. 13 Energy Saving Swing, thus allow speed to increase to 80km/h
- Around km. 11.5 take rheostat brakes to control speed
- After Umbulwana, before km. 7.5 speed must be controlled at 50km/h, to adhere to curve speed restriction,
- Around km. 7 take traction to control train speed
- After km. 6.5 speed must be controlled at 60km/h, to adhere to curve speed restriction,
- Enter Ladysmith at 30km/h, as driving over steel bridge

Ladysmith To Vooruitsig

- Exit Ladysmith with train speed of 30km/h with traction
- Pass through Danskraal South at 30km/h
- Before km. 181 exit Danskraal North at 30km/h
- From km. 180, allow speed to increase and control speed at 60km/h
- Around km. 171.5 take rheostat to control speed
- Around km. 170 Energy Saving Swing, thus allow train to swing to 80km/h
- Around km. 168.5 take traction to control speed at 60km/h
- From km. 163 decrease traction and switch over to rheostat brakes
- After km. 161.5, speed must be controlled at 60km/h, to adhere to curve speed restriction,
- Enter Elandslaagte at 50km/h and exit at 60km/h
- From km. 160 take traction and allow speed to increase and control at 60km/h
- Around km. 158 take rheo if needed to control speed at 60km/h
- Around km. 156.5 switch back to traction and control speed
- *From km. 147.5 speed must be controlled at 50km/h, to adhere to down gradient speed restriction, up to km. 141.5*
- Around km. 147.5 take rheostat and apply train brakes if and when needed to control speed

- Around km. 141.5 Energy Saving Swing, thus allow train to swing to 80km/h
- From km. 140 take traction in Wasbank and maintain speed at 60km/h
- Around km. 138 change to rheostat to control speed, allowing train to swing to 80km/h
- From km. 136.5 take traction to maintain speed at 60km/h
- After km. 129 speed must be controlled at 60km/h, to adhere to curve speed restrictions,
- Enter Glencoe at 30km/h, around km. 121.5 and exit also at 30km/h, since diamond crossing present
- Allow speed to increase and control speed at 60km/h
- Around km. 116.5 take rheostat and control speed at 60km/h
- Around km. 112 take traction to control speed at 60km/h
- From km. 102 decrease traction and take rheostat to control speed
- *After Danhauser, from km. 96 speed must be controlled at 40km/h, to adhere to down gradient speed restriction, up to km. 91.5*
- Apply train brakes when and as required to control train speed at 60km/h
- *After Milnerdale, from km. 90 speed must be controlled at 50km/h, to adhere to down gradient speed restriction, up to km. 85.5*
- Around km. 85 release train brakes and allow speed to increase and control speed at 60km/h
- Around km. 82 take traction to control speed at 60km/h
- At km. 79 exit Ballengeich at 30km/h
- Km. 72.5 close traction off to M1 and allow train to swing at 80km/h
- Around km. 71 increase traction to maintain speed
- Around km. 63 take rheostat to control speed at 60km/h
- *After Newcastle, from km. 62.5 speed must be controlled at 50km/h, to adhere to down gradient speed restriction, up to km. 60*
- Apply train brakes if and as needed
- Around km. 58.5 take traction to maintain speed
- From km. 54 decrease traction and take rheostat to control speed at 50km/h, to *adhere to down gradient speed restriction, up to km. 52*

- Around km. 49 take traction to control speed at 60km/h
- From km. 9 decrease traction and take rheostat braking as needed to control speed at 60km/h
- Around km. 4.5 take traction to maintain speed
- After km. 0.5 switch to rheostat when needed to control speed
- Enter Vooruitsig at 20km/h

Vooruitsig To Union

- Allow speed to increase and control speed at 60km/h with rheostat
- Around km. 222.5 take traction to maintain speed
- Around km. 219.5 decrease traction and take rheostat to control train speed
- Around km. 216 take traction to control speed at 60km/h
- From km. 200 decrease traction and change to rheostat to control speed at 60km/h
- Around km. 187.5 take traction to maintain speed
- Around km. 184.5 take rheostat to control speed
- *From km. 183 speed must be controlled at 50km/h, to adhere to down gradient speed restriction, up to km. 181*
- Apply train brakes if and as needed to control train speed
- Around km. 179.5 take traction if needed to control speed
- From km. 178 control speed with rheostat braking at 60km/h
- Around km. 158 reduce rheostat and take traction to maintain speed
- Around km. 156 take rheostat to control speed
- *From km. 152.5 speed must be controlled at 50km/h, to adhere to down gradient speed restriction, up to km. 150*
- Apply train brakes if and as needed to control train speed
- Around km. 149 take traction to maintain speed and allow speed to increase to 60km/h
- Around km. 143 reduce traction and change to rheostat to control speed at 60km/h
- **Km. 142 Energy Saving Swing?**
- Around km. 140 take traction to control speed through Elmtree

- Around km. 131 decrease traction and change to rheostat to control speed at 60km/h
- Around km. 127 take traction to maintain speed
- Around km. 125 take rheostat to control speed at 60km/h
- From km. 117 take traction to maintain speed
- Around km. 107 take rheostat to control speed at 60km/h
- Around km. 101.5 take traction to maintain speed
- From km. 96 decrease traction and change to rheostat to control speed at 60km/h
- **Km. 94.5 Energy Saving Swing?**
- Around km. 93 take traction to maintain speed of 60km/h
- From around km. 78 decrease traction and take rheostat to control speed at 60km/h
- *From km. 75, just before Balfour, speed must be controlled at 50km/h, to adhere to down gradient speed restriction, up to km. 73*
- Apply train brakes if and as needed to control speed
- From km. 62 take traction to maintain speed
- Around km. 53 change to rheostat to control speed at 60km/h
- *From km. 52, speed must be controlled at 50km/h, to adhere to down gradient speed restriction, up to km. 50*
- *After Kraal, from km. 47.5, speed must be controlled at 50km/h, to adhere to down gradient speed restriction, up to km. 45*
- Apply train brakes when and as needed
- Around km. 42.5 release train brakes
- After km. 42 take traction to maintain speed at 60km/h
- Around km. 22 decrease traction and change to rheostat to control speed
- **Km. 18 Energy Saving Swing?**
- From km. 17 decrease rheostat and change to traction to maintain speed
- Around km. 11 take rheostat to control speed
- Pass through Rooikop at 30km/h to confirm signals and cross over to mainline
- Before km. 4 speed must be controlled at 60km/h, to adhere to curve speed restriction
- Around km. 3.5 take traction to control speed
- Control train speed through Union station at 30km/h for diamond crossing

Union To City Deep

- Control speed with traction past Refinery
- Use rheostatic brakes to control speed into City Deep

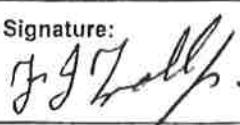
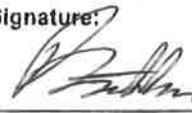
TRANSNET FREIGHT RAIL			
File Ref: S.RTS 1/6/16/8	LOADING SPECIFICATION FOR SECURING AND TRANSPORTING OF CONTAINERS		Document no: RSE/TE/LPR/0040
			Revision: 2
Author:	Mr F J Trollip Manager Transnet Engineering	Signature: 	Issue date: 26 July 2016
Technical Approving Officer:	Dr R D Fröhling Principal Engineer Technology Management	Signature: 	Page 1 of 8
Operational Approving Officer:	Mr L Nel Manager Loading Specifications	Signature: 	

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SCHEDULE OF AMENDMENTS

Revision number	Section amended	Summary of amendments	Date issued
2	-	Unsymmetrical loading of containers and the transportation of explosives has been amended/added. Subsequently some sections have been renumbered.	26 July 2016

1. INTRODUCTION

This loading specification is for the securing and transporting of containers on rail wagons.

All local designed containers must first be certified by Transnet, Technology Management, before the containers can be transported by rail.

This loading specification supersedes all previous loading instructions, letters, granted concessions and granted deviations. It is the responsibility of the recipient to destroy all old documents.

Wagons not loaded according to this specification, will not be accepted for conveyance. See transport agreement for actions if wagons are not loaded correctly.

2. PRODUCT DESCRIPTION

ISO containers, 20' and 40'

Reefer container

Tanktainers

Flat racks

Local designed containers, 3, 6 and 12 m

The containers will be loaded with a variety of goods to be exported, imported or for local distribution in closed or open top containers.



20' Dry container



Reefer container



Tanktainer



Flat rack

3. CONTAINERS

- Locally used containers must be certified according to Transnet specification BBF 8110.
- Before a container is loaded, ensure that the container adheres to all national/international standards and that the doors and their locking mechanism are in a working condition.
- The correct load distribution and securing of the load in the container is the responsibility of the customer loading the container.
- Tanktainers must be loaded below 20% or above 80% of its capacity, no loading between 20 and 80% are allowed.
- Containers loaded with dangerous goods must conform to all relevant UN loading regulations.
- Open top containers and flat racks, with goods protruding above the top edge, must get special permission from the loading specification office before it can be transported by rail.

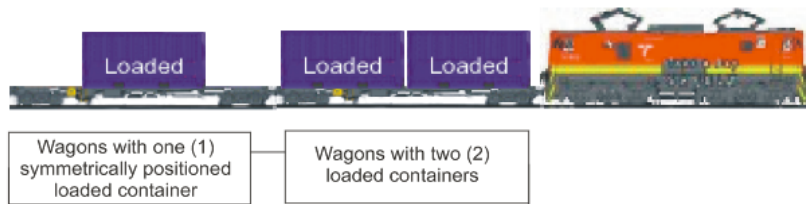
4. WAGON REQUIREMENTS

- Before any wagon is loaded, the wagon must be clean and in a rail-worthy condition.
- All foreign objects must be removed from the wagon.
- Under no circumstances shall defective equipment or wagons be used.
- Wagons with defective twist locks shall not be used.
- Wagons that are technically suited to transport containers are SHJ, SHLJ, SHR, SMJ, SMLJ and SMR types of wagons.

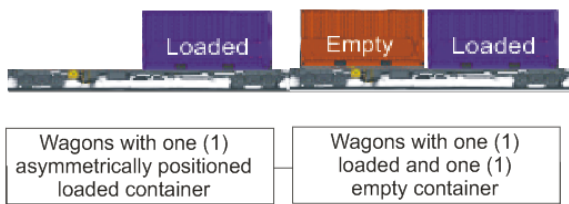
5. PAYLOAD DISTRIBUTION AND SAFETY LIMITS FOR WAGONS

- The carrying capacity ("Load") stencilled on the wagon shall not be exceeded. No overloading is permitted.
 - o Wagons coded SMLJ or SHLJ have a maximum payload of 48 000 kg.
 - o Wagons coded SMR or SHR have a maximum payload of 61 000 kg.
- The maximum gross mass of a 6 m and a 12 m (20' and 40') ISO container is 30 480 kg.
- If the payload of the wagon is 48 000 kg only one 6 m ISO container with a gross mass of 30 480 kg may be loaded.
- If the payload of the wagon is 61 000 kg two 6 m ISO containers with a gross mass of 30 480 kg each may be loaded on the wagon.
- The load in the container shall be positioned such that, as far as possible, the load is evenly distributed transversely (from side to side) and longitudinally (from end to end).
- The maximum allowable Side-to-Side skew loading limit of the wagon is 5 ton.
- It is preferred that two 6 m containers are placed door to door on a wagon.
- When a single 6 m general freight container is placed on a wagon it is preferred to position it next to a bathtub bracket for security reasons.
- Any combination of loaded and empty containers can be loaded on a single wagon, but the container train compilation rules as shown below and as defined in Instructions S.GFB/TES/TE/OI 257 and S.GFB/TES/TE/OI 0334 shall be adhered to.

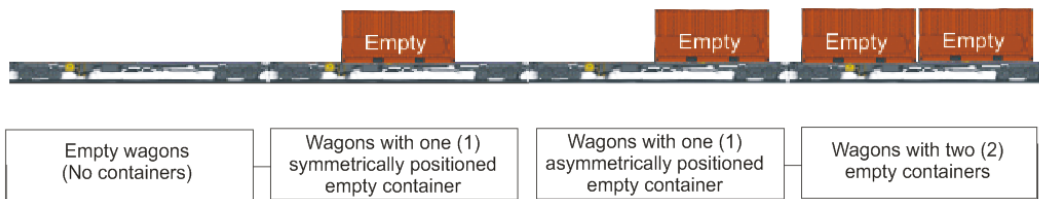
Front part of the train



Middle part of the train



Back part of the train



6. TRANSPORTING OF EXPLOSIVES IN CONTAINERS

- Containers containing explosives must be loaded door to door
- Only wagons fitted with friction draw gear (short buffer) are suitable to transport explosives. No centre sliding sill or end of car cushioning device draw gear wagons shall be used
- 2 Loaded container wagons, carrying loaded or empty containers, (non-explosives), must be used as runners behind the locos
- 2 empty wagons must be used as runners at the back of the train
- Single 20ft explosives containers must be centrally loaded and a SARU blocker loaded adjacent to prevent the doors from being tampered with. If no blocker is available locking devices must be fitted
- For more information regarding the transportation of explosives in containers see loading specification RSE/TE/LPR/0034 and General Appendix No 6

7. TWIST LOCKS FITTED TO RAIL WAGONS

Transnet wagons are fitted with the following type of twist locks:

- Retractable twist locks, see Photo 1
- Foldable twist lock, see Photo 2



Photo 1

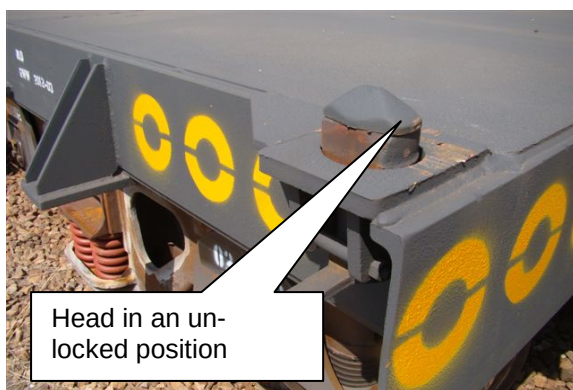


Photo 2

8. LOADING AND SECURING OF CONTAINERS ON RAIL WAGONS

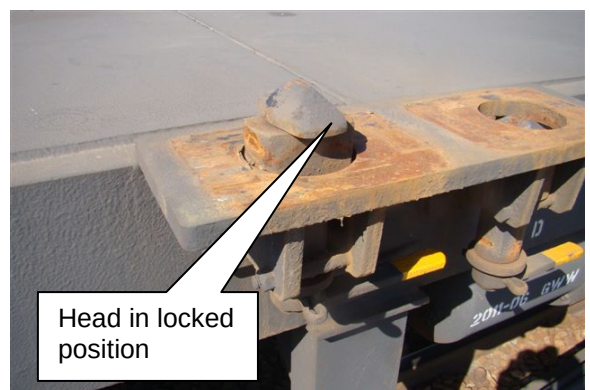
- Ensure that the doors of the containers are closed and properly locked before the containers are loaded onto a wagon
- It is preferred that the containers are loaded door to door on the rail wagon
- If the containers cannot be loaded door to door the doors must be locked with an appropriate lock agreed by all parties
- Each container loaded on a container wagon must be secured with 4 twist locks provided on the wagon, before departure from the terminal

8.1. Operating Procedure for Retractable Twist Locks



Head in an un-
locked position

Photo 3 - Corner twist lock



Head in locked
position

Photo 4 - Middle twist lock

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Shank must be retracted if not used to secure container

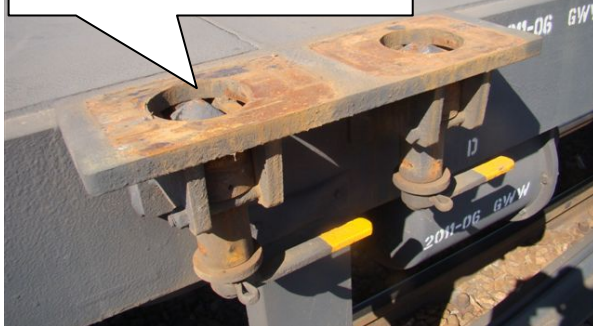


Photo 5 - Middle twist locks

Shank must be lifted and un-locked to accommodate container



Photo 6 - Shank lifted for loading

Turned handle to this position to lock container in position



Photo 7 - Middle twist lock locked

Turn handle to this position to un-lock container

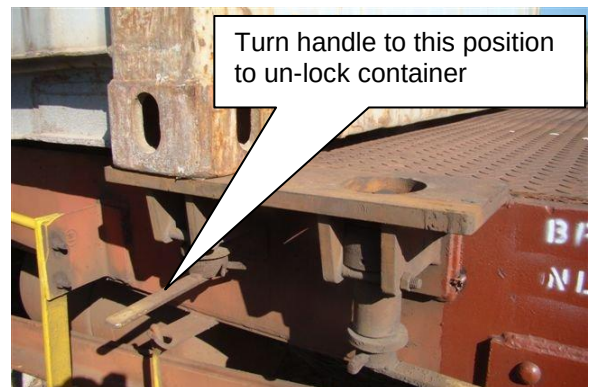


Photo 8 - Container un-locked

8.2. Operating Procedure for Foldable Twist Locks

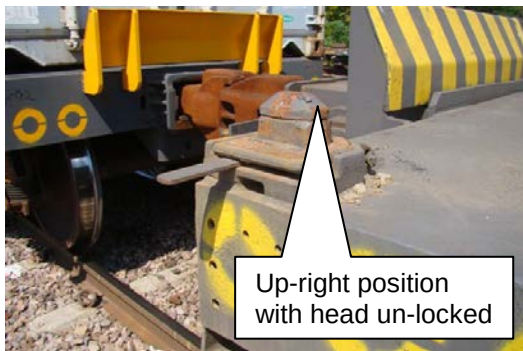


Photo 9 - Corner twist lock

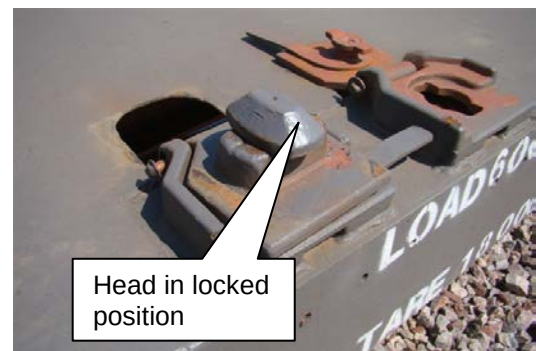


Photo 10 - Middle twist lock

Twist locks folded away if not used to secure container



Photo 11 - Middle twist locks

Lifted twist lock in an un-locked position to accommodate container



Photo 12 - Twist lock in loading position

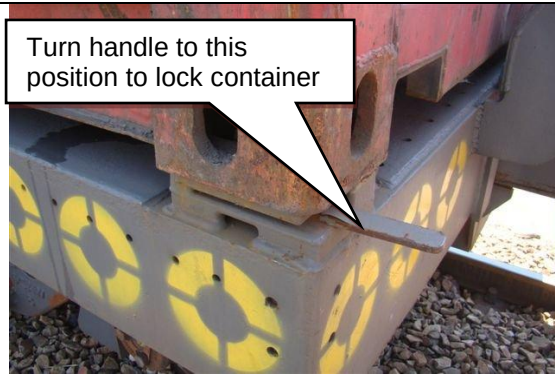


Photo 13 – Handle in locked position

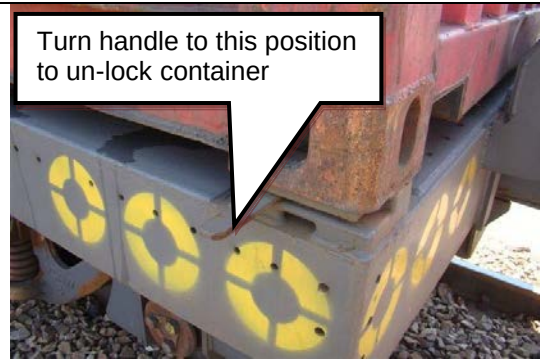


Photo 14 – handle in un-locked position

9. CUSTOMER CO-OPERATION

- The customer must adhere to the loading specification. Any deviations to the loading method must be requested in writing to the Account Manager and the Product Development department in Transnet Engineering
- Transnet Freight Rail must be informed if the loading/off-loading methods and/or product have changed
- The customer may not perform any modification of any nature to Transnet Freight Rail wagons. If there is a need for modifications, the Technical Fleet Owner must approve the request and arrange for the necessary modifications.

10. OPENING OF CONTAINERS

Adequate precautions shall be taken when opening the doors not to get injured as a result of possible load shifting during transit

11. AMENDMENT OF LOADING SPECIFICATION

This loading specification is subject to change without prior notice if it either impacts on safety or if modifications/improvements are necessary

12. REFERENCES

<http://www.evansdist.com/blog/2013/09/types-of-shipping-containers/>

Transnet specification S.GFB/TES/TE/OI 257

Transnet specification S.GFB/TES/TE/OI 0334

Loading specification RSE/TE/LPR/0034

General Appendix No 6